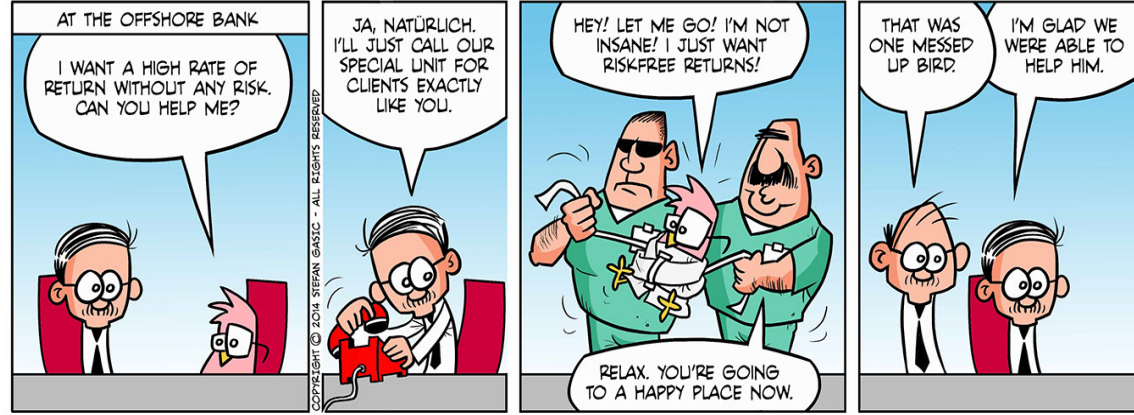


Portfolio Management Project

Michaela Shoults



The Decision



Goal: Decide which risky securities to invest in for the most profitable pension scheme.

- Compared to investing a pension fund into a risk-free bank account, investing in risky securities in addition to a bank account could lead to the highest expected return.



Case Overview

- You retire after 40 years and want to receive pension for 20 years.
- Your salary grew 2% annually. Pension starts at 50% of FINAL salary and grows 2% annually.
- Interest rate offered by risk-free account is 5%
- You consider a risk of up to 10% to be reasonable.
- Three stocks are selected (A, B, C), which are included in your portfolio with your bank account.
- The annual expected returns and standard deviations for each stock can be estimated. Correlation coefficients between the stocks can also be estimated.
- The three chosen stocks constitute a mini-market, so you must select a suitable portfolio on the capital market line (CML)
- As a small investor, short selling is likely not possible, so portfolios need to be restricted to non-negative weights.



Procedure

- Use Solver in Excel to maximise the gradient of the straight line through the risk-free security and feasible portfolios to find non-negative weights in market portfolio M'
- Solver gives the optimal risky portfolio M' (weights, Expected return on M', and risk on M')
- Combine M' with risk-free investment into 5% interest account to construct portfolio P on the CML
- The choice of how to split the pension fund is now based on your attitude towards risk.
- Example:
 - To construct a portfolio P with 10% risk, 16.19% of funds go into bank account while rest go into M'. The expected return on P is then found to be 9.7%.
- Compare expected return for 0% risk to expected return for X% risk
- The expected return for risk greater than 0% is always higher than expected return for 0% risk.

Application

- Want to retire after 40 years, 20 years pension, salary+2% per year, pension+2% per year. Pension starts at 50% of final salary.
- Interest rate for risk-free account is 5%.
- Salary is \$100,000/year
 - Geometric growth formula: $100,000(1+0.02)^{40} = \$220,803.97$ final salary
 - Pension starts at \$110,401.98



Underperformance



Risk-adjusted
returns

Risky Assets



- Chosen stocks: Apple Inc. (AAPL), [Amazon.com](https://www.amazon.com), Inc. (AMZN), Netflix, Inc. (NFLX)
- Data from the past six years, with a daily frequency, of each stock was downloaded from Google Finance into Google Sheets, then transferred to Excel.
- The return rate is calculated for each stock from the given prices

$$k = \frac{S(T) - S(0)}{S(0)}$$

- Expected return values and Std. Deviations are converted from daily to annual using 252 as a multiplier. (252 trading days in a year)

AAPL			AMZN			NFLX		
Date	Close	Return	Date	Close	Return	Date	Close	Return
1/2/20 16:00	75.09	-0.00972	#####	94.9	-0.01212	1/2/20 16:00	32.98	-0.01183
1/3/20 16:00	74.36	0.007934	#####	93.75	0.014827	1/3/20 16:00	32.59	0.030377
1/6/20 16:00	74.95	-0.00467	#####	95.14	0.002102	1/6/20 16:00	33.58	-0.01489
1/7/20 16:00	74.6	0.016086	#####	95.34	-0.00776	1/7/20 16:00	33.08	0.025695
1/8/20 16:00	75.8	0.02124	#####	94.6	0.004757	1/8/20 16:00	33.93	-0.01061
1/9/20 16:00	77.41	0.002196	#####	95.05	-0.00936	1/9/20 16:00	33.57	-0.01966
1/10/20 16:00	77.58	0.021397	#####	94.16	0.004354	1/10/20 16:00	32.91	0.029778
1/13/20 16:00	79.24	-0.0135	#####	94.57	-0.01163	1/13/20 16:00	33.89	-0.00059
1/14/20 16:00	78.17	-0.00435	#####	93.47	-0.00396	1/14/20 16:00	33.87	0.001181
1/15/20 16:00	77.83	0.012592	#####	93.1	0.008593	1/15/20 16:00	33.91	-0.00147
1/16/20 16:00	78.81	0.011039	#####	93.9	-0.00703	1/16/20 16:00	33.86	0.003249
1/17/20 16:00	79.68	-0.00678	#####	93.24	0.014586	1/17/20 16:00	33.97	-0.00471
1/21/20 16:00	79.14	0.003664	#####	94.6	-0.00243	1/21/20 16:00	33.81	-0.03579
1/22/20 16:00	79.43	0.004784	#####	94.37	-0.00148	1/22/20 16:00	32.6	0.072393
1/23/20 16:00	79.81	-0.00288	#####	94.23	-0.0122	1/23/20 16:00	34.96	0.010297
1/24/20 16:00	79.58	-0.0294	#####	93.08	-0.01783	1/24/20 16:00	35.32	-0.02916
1/27/20 16:00	77.24	0.028224	#####	91.42	0.013564	1/27/20 16:00	34.29	0.016331
1/28/20 16:00	79.42	0.020902	#####	92.66	0.00259	1/28/20 16:00	34.85	-0.01521
1/29/20 16:00	81.08	-0.00136	#####	92.9	0.006781	1/29/20 16:00	34.32	0.013112
1/30/20 16:00	80.97	-0.04434	#####	93.53	0.07388	1/30/20 16:00	34.77	-0.00748
1/31/20 16:00	77.38	-0.00271	#####	100.44	-0.00229	1/31/20 16:00	34.51	0.03738
2/3/20 16:00	77.17	0.032914	#####	100.21	0.022652	2/3/20 16:00	35.8	0.030726
2/4/20 16:00	79.71	0.008155	#####	102.48	-0.00478	2/4/20 16:00	36.9	0.001897



Average Returns, Std. Deviation, & Correlation

Using Excel (AAPL):

	AAPL	AMZN	NFLX
Expected Return	0.091793	0.058009	0.103658
Std. Dev.	0.508695	0.534529	0.592154

- `=(AVERAGE(C14:C1596))*252`
- `=(STDEV.S(C14:C1596))*SQRT(252)`

Correlation Matrix:		AAPL	AMZN	NFLX
	AAPL	1	0.823639	0.723051
	AMZN	0.823639	1	0.753024
	NFLX	0.723051	0.753024	1

`=CORREL(array of AAPL return rates, array of AMZN/NFLX return rates)`



Covariance Matrix

Covariance Matrix:		AAPL	AMZN	NFLX
	AAPL	0.258771	0.223958	0.217802
	AMZN	0.223958	0.285721	0.23835
	NFLX	0.217802	0.23835	0.350647

$$Cov(i, j) = \rho_{ij}\sigma_i\sigma_j \quad \text{where } \rho_{ij} \text{ is the correlation coefficient}$$

In Excel (AAPL & AMZN):

- =(correlation coeff. between AAPL/AMZN)*(Std. Dev. of AAPL)*(Std. Dev. of AMZN)



Risky Portfolio

- Initial weights are established so Solver in Excel can have a starting point
 - An equal amount of money is invested.

- Return rate:

$$K_v = w_1\mu_1 + w_2\mu_2 + w_3\mu_3$$

- Slope of CML:

$$\frac{K_v - 0.05}{\sigma_v}$$

- Risk:

$$\sigma_v = \sqrt{w_1^2\mu_1^2 + w_2^2\mu_2^2 + w_3^2\mu_3^2 + 2w_1w_2(Cov_{1,2}) + 2w_1w_3(Cov_{1,3}) + 2w_2w_3(Cov_{2,3})}$$

Weight AAPL:	0.33
Weight AMZN:	0.33
Weight NFLX:	0.34

E[Return] on P:	0.084678
Portfolio Risk:	0.39788
Slope of CML:	0.087157

Portfolio Optimization

- Use Solver in Excel to create an optimal risky market portfolio M'
- Constraints:
 - Weights must add to be equal to 1
 - Weights must be greater than 0 (no short selling)
- Objective: CML slope
- Changing variables: weights
- Solver decided Netflix gave the best risk-adjusted return compared to any combination

Weight AAPL:	0	E[Return] on P:	0.103658
Weight AMZN:	0	Portfolio Risk:	0.103658
Weight NFLX:	1	Slope of CML:	0.517645

Solver Parameters

Set Objective:

To: ☒ Max ☐ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

\$M\$17 = 1

\$N\$8:\$N\$10 >= 0

Add
Change
Delete
Reset All
Load/Save

☒ Make Unconstrained Variables Non-Negative

Select a Solving Method:

Options

Solving Method

Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Close

Solve



Portfolio P on CML

- Combine M' with risk-free investment bank account at 5% interest to make portfolio P on the CML
- The amount of money invested is dependent on attitude toward the risk
- w is the amount of money chosen to invest in M'
- $(1 - w)$ is the amount of money invest in bank account

$$\sigma_P = w\sigma_{M'}$$

$$w = \frac{\sigma_P}{\sigma_{M'}}$$

$$E[R_P] = 0.05 + \frac{E[R_{M'}] - 0.05}{\sigma_{M'}} \sigma_P$$

$$E[R_P] = w(E[R_{M'}]) + (1 - w)(0.05)$$



Risk vs. Return Table

<u>Risk</u>	<u>0.00%</u>	<u>2.50%</u>	<u>5.00%</u>	<u>7.50%</u>	<u>10.00%</u>
<u>Expected Return</u>	5%	6.29%	7.59%	8.88%	10.18%
<u>Portion of Salary Paid</u>	0%	24.12%	48.24%	72.35%	96.47%
<u>Of Which Invested Risk Free</u>	100%	75.88%	51.76%	27.65%	3.53%

Conclusion



- High risk → High investment → High return
- Low risk → Low investment → Low return
- If we invested into a Netflix stock with 5% risk (7.59% return) from our first pension salary:
 - 48.24% of \$110,401.98: \$53,257.92
 - Return: \$4,042.28
- Pension after one year with no-risk investment: \$118,130.12
- Pension after one year with 5% risk investment: \$118,787.50

The image features a solid light pink background. In the top-left corner, there are three vertical bars of varying heights, each composed of four overlapping circles. In the bottom-right corner, there are four vertical bars of varying heights, each composed of four overlapping circles. The word "Questions?" is centered in the middle of the image in a bold, dark pink font.

Questions?



References

- Yuan Zhou Lecture Slides:
https://lucky-my.sharepoint.com/personal/yzh392_uky_edu/_layouts/15/Doc.aspx?sourcedoc={e4e45b00-ee25-4b5d-bb94-e473a4490d86}&action=view&wd=target%28Notes.one%7C86379a56-b702-5d4c-9db0-bd2779a616ec%2F13.%20Portfolio%20on%20several%20securities%2C%20MVP%7C74940714-17fb-b747-9f90-d248cd457a73%2F%29&wdorigin=NavigationUrl
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